

ANALYSIS 1

PREPARATION FOR TEST 2

Exercise 1. Evaluate the following sum:

$$\sum_{n=1}^{+\infty} \frac{2^{n+1} \cdot 3^n + (-3)^{n+1}}{3^{2n+1}}.$$

Solution By the transformation

$$\frac{2^{n+1} \cdot 3^n + (-3)^{n+1}}{3^{2n+1}} = \frac{2 \cdot 2^n \cdot 3^n - 3 \cdot (-3)^n}{3 \cdot 9^n} = \frac{2}{3} \cdot \left(\frac{2}{3}\right)^n - \left(-\frac{1}{3}\right)^n,$$

we can see that the series is the linear combination of geometric series with ratios $q = \frac{2}{3}$ and $q = -\frac{1}{3}$. In both cases $|q| < 1$, so the series is convergent and its sum is

$$\begin{aligned} \sum_{n=1}^{+\infty} \frac{2^{n+1} \cdot 3^n + (-3)^{n+1}}{3^{2n+1}} &= \frac{2}{3} \sum_{n=1}^{+\infty} \left(\frac{2}{3}\right)^n - \sum_{n=1}^{+\infty} \left(-\frac{1}{3}\right)^n = \frac{2}{3} \sum_{n=0}^{+\infty} \left(\frac{2}{3}\right)^{n+1} - \sum_{n=0}^{+\infty} \left(-\frac{1}{3}\right)^{n+1} \\ &= \frac{2}{3} \sum_{n=0}^{+\infty} \left(\frac{2}{3} \left(\frac{2}{3}\right)^n\right) - \sum_{n=0}^{+\infty} \left(\left(-\frac{1}{3}\right) \left(-\frac{1}{3}\right)^n\right) \\ &= \frac{4}{9} \sum_{n=0}^{+\infty} \left(\frac{2}{3}\right)^n + \frac{1}{3} \sum_{n=0}^{+\infty} \left(-\frac{1}{3}\right)^n = \frac{4}{9} \cdot \frac{1}{1 - \frac{2}{3}} + \frac{1}{3} \cdot \frac{1}{1 + \frac{1}{3}} = \underline{\underline{\frac{19}{12}}}. \end{aligned}$$

Exercise 2. Determine whether the following series are convergent or divergent. Prove your statements.

$$a) \sum_{n=1} \frac{2n-1}{\sqrt[3]{n^6 + 7n^2 - 1}}, \quad b) \sum_{n=0} \left(\frac{3n-1}{3n+5}\right)^{n^2+1}, \quad c) \sum_{n=0} \frac{(2n+1)!}{(n!)^2 \cdot 3^{n+2}}.$$

Solution

a) We guess that the series diverges, since

$$\frac{2n-1}{\sqrt[3]{n^6 + 7n^2 - 1}} \approx \frac{2n}{\sqrt[3]{n^6}} = \frac{2}{n}, \quad \text{if } n \text{ is sufficiently large, and } \sum_{n=1} \frac{1}{n} \text{ diverges.}$$

To prove this, we find a divergent minorant series. By the estimations

$$2n-1 = n + (n-1) \underset{(n \geq 1)}{\geq} n + 0 = n$$

and

$$\sqrt[3]{n^6 + 7n^2 - 1} \leq \sqrt[3]{n^6 + 7n^2} \leq \sqrt[3]{n^6 + 7n^6} = \sqrt[3]{8n^6} = 2n^2$$

we have

$$a_n := \frac{2n-1}{\sqrt[3]{n^6 + 7n^2 - 1}} \underset{(n \geq 1)}{\geq} \frac{n}{2n^2} = \frac{1}{2n} =: b_n.$$

According to the divergence of the harmonic series $\sum_{n=1} b_n$ is divergent. Then, by minorant criterion, $\sum_{n=1} a_n$ is also divergent.

b) By Cauchy root test, the series is convergent. Indeed,

$$\begin{aligned} a_n = \left(\frac{3n-1}{3n+5} \right)^{n^2+1} &\implies \sqrt[n]{|a_n|} = \left(\frac{3n-1}{3n+5} \right)^{n+1/n} = \left(\frac{3n-1}{3n+5} \right)^n \cdot \sqrt[n]{\frac{3n-1}{3n+5}} \\ &\xrightarrow{n \rightarrow +\infty} \frac{1}{e^2} \cdot 1 = \frac{1}{e^2} < 1, \end{aligned}$$

since

$$\left(\frac{3n-1}{3n+5} \right)^n = \frac{\left(1 + \frac{-1/3}{n} \right)^n}{\left(1 + \frac{5/3}{n} \right)^n} \xrightarrow{n \rightarrow +\infty} \frac{e^{-1/3}}{e^{5/3}} = \frac{1}{e^2}$$

and

$$x_n := \frac{3n-1}{3n+5} = \frac{3 - \frac{1}{n}}{3 + \frac{5}{n}} \xrightarrow{n \rightarrow +\infty} \frac{3-0}{3+0} = 1 \in (0, +\infty) \implies \sqrt[n]{x_n} \xrightarrow{n \rightarrow +\infty} 1.$$

c) By d'Alembert's ratio test, the series is divergent. Indeed,

$$\begin{aligned} a_n = \frac{(2n+1)!}{(n!)^2 \cdot 3^{n+2}} &\implies \left| \frac{a_{n+1}}{a_n} \right| = \frac{(2(n+1)+1)!}{((n+1)!)^2 \cdot 3^{(n+1)+2}} \cdot \frac{(n!)^2 \cdot 3^{n+2}}{(2n+1)!} \\ &= \frac{(2n+3)!}{((n+1) \cdot n!)^2 \cdot 3^{n+3}} \cdot \frac{(n!)^2 \cdot 3^{n+2}}{(2n+1)!} \\ &= \frac{(2n+3)(2n+2) \cdot (2n+1)!}{(n+1)^2 \cdot (n!)^2 \cdot 3 \cdot 3^{n+2}} \cdot \frac{(n!)^2 \cdot 3^{n+2}}{(2n+1)!} \\ &= \frac{(2n+3)(2n+2)}{3(n+1)^2} = \frac{(2 + \frac{3}{n})(2 + \frac{2}{n})}{3(1 + \frac{1}{n})^2} \\ &\xrightarrow{n \rightarrow +\infty} \frac{(2+0)(2+0)}{3(1+0)^2} = \frac{4}{3} > 1. \end{aligned}$$

Exercise 3. Determine the convergence set of the following power series:

$$\sum_{n=0} \frac{2^n}{n^3 + 1} \cdot (x-1)^n \quad (x \in \mathbb{R}).$$

Solution The center of the power series is $a = 1$. By calculation based on roots:

$$\alpha_n = \frac{2^n}{n^3 + 1} \quad \Rightarrow \quad \sqrt[n]{|\alpha_n|} = \frac{2}{\sqrt[n]{n^3 + 1}} = \frac{2}{(\sqrt[n]{n})^3 \cdot \sqrt[n]{1 + \frac{1}{n^3}}} \xrightarrow{n \rightarrow +\infty} \frac{2}{1^3 \cdot 1} = 2 = A,$$

which implies that $R = \frac{1}{A} = \frac{1}{2}$ and $\mathfrak{C}^\circ = (1 - \frac{1}{2}, 1 + \frac{1}{2}) = (\frac{1}{2}, \frac{3}{2})$.

Analysis of the boundary points:

- For $x = \frac{3}{2}$ we have the series

$$\sum_{n=0}^{\infty} \frac{2^n}{n^3 + 1} \cdot \left(\frac{3}{2} - 1\right)^n = \sum_{n=0}^{\infty} \frac{2^n}{n^3 + 1} \cdot \left(\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{n^3 + 1}$$

which is convergent, since by Majorant Criterion

$$0 \leq \frac{1}{n^3 + 1} \leq \frac{1}{n^3} := b_n,$$

and according to the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$, when $\alpha > 1$, the series $\sum_{n=1}^{\infty} b_n$ is convergent.

- For $x = \frac{1}{2}$ we have the series

$$\sum_{n=0}^{\infty} \frac{2^n}{n^3 + 1} \cdot \left(\frac{1}{2} - 1\right)^n = \sum_{n=0}^{\infty} \frac{2^n}{n^3 + 1} \cdot \left(-\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n^3 + 1}$$

which is convergent, since it is absolute convergent (see the case for $x = \frac{3}{2}$).

In summary $\mathfrak{C} = \left[\frac{1}{2}, \frac{3}{2}\right]$.

Exercise 4. Evaluate the following limit, and prove by definition your result:

$$\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 + 3x}{x^2 - x - 2}$$

Solution First we evaluate the limit:

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 + 4x^2 + 3x}{x^2 - x - 2} &= \left(\frac{0}{0}\right) = \lim_{x \rightarrow -1} \frac{x(x^2 + 4x + 3)}{x^2 - x - 2} = \lim_{x \rightarrow -1} \frac{x \cancel{(x+1)}(x+3)}{\cancel{(x+1)}(x-2)} \\ &= \frac{-1 \cdot (-1+3)}{-1-2} = \frac{2}{-3} = -\frac{2}{3}. \end{aligned}$$

Now, by the definition of limit, using the transformation

$$\frac{x^3 + 4x^2 + 3x}{x^2 - x - 2} = \frac{x(x+3)}{x-2} \quad (x \in \mathbb{R} \setminus \{-1, 2\}),$$

we have to prove that

$$(*) \quad \forall \varepsilon > 0 \exists \delta > 0, \forall x \in \mathbb{R} \setminus \{-1, 2\}, 0 < |x + 1| < \delta: \left| \frac{x(x+3)}{x-2} - \frac{2}{3} \right| < \varepsilon.$$

Let $\varepsilon > 0$ be a fixed real number. Then, for all $x \in \mathbb{R} \setminus \{-1, 2\}$,

$$\begin{aligned} \left| \frac{x(x+3)}{x-2} - \frac{2}{3} \right| &= \left| \frac{3x(x+3) - 2(x-2)}{3(x-2)} \right| = \left| \frac{3x^2 + 7x + 4}{3(x-2)} \right| = \left| \frac{(x+1)(3x+4)}{3(x-2)} \right| \\ &= \left| \frac{3x+4}{3(x-2)} \right| \cdot |x+1| \leq \frac{|3x|+4}{3|x-2|} \cdot |x+1| = \frac{3|x|+4}{3|x-2|} \cdot |x+1|. \end{aligned}$$

The condition $|x+1| < \delta$ means that we are in a neighborhood around -1 with radio δ . We take the restriction $\delta < 1$, that is, $|x+1| < 1$. Then,

$$-1 < x+1 < 1 \implies -2 < x < 0 \implies 0 < |x| < 2$$

and

$$-1 < x+1 < 1 \implies -4 < x-2 < -2 \implies 2 < |x-2| < 4.$$

Thus,

$$\left| \frac{x(x+3)}{x-2} - \frac{2}{3} \right| \leq \frac{3|x|+4}{3|x-2|} \cdot |x+1| \leq \frac{3 \cdot 2 + 4}{3 \cdot 2} \cdot |x+1| = \frac{10}{6} \cdot |x+1| \leq \underbrace{2|x+1|}_{|x+1| < \frac{\varepsilon}{2}} < \varepsilon.$$

Hence, with the choice $\delta := \min\left\{1, \frac{\varepsilon}{2}\right\}$, we proved that $(*)$ is satisfied.

Exercise 5. Evaluate the following limits (do not use L'Hospital's rule here):

$$a) \lim_{x \rightarrow 0} \frac{\sqrt{x} \cdot \sin(\sqrt{x})}{\sqrt{2x+4}-2}, \quad b) \lim_{x \rightarrow 0} \frac{x(e^{4x} - e^{2x})}{1 - \cos^3 x}.$$

Solution

a) We use the following transformation:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x} \cdot \sin(\sqrt{x})}{\sqrt{2x+4}-2} = \lim_{x \rightarrow 0} \frac{\sin(\sqrt{x})}{\sqrt{x}} \cdot \frac{x}{\sqrt{2x+4}-2} = \lim_{x \rightarrow 0} \frac{\sin(\sqrt{x})}{\sqrt{x}} \cdot \lim_{x \rightarrow 0} \frac{x}{\sqrt{2x+4}-2}.$$

By the substitution $y = \sqrt{x} \rightarrow 0$ if $x \rightarrow 0$ we have

$$\lim_{x \rightarrow 0} \frac{\sin(\sqrt{x})}{\sqrt{x}} = \lim_{y \rightarrow 0} \frac{\sin(y)}{y} = 1.$$

Moreover, dividing and multiplying by the conjugate of the denominator, we obtain

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x}{\sqrt{2x+4}-2} &= \lim_{x \rightarrow 0} \frac{x(\sqrt{2x+4}+2)}{(2x+4)-2^2} \lim_{x \rightarrow 0} \frac{\cancel{x}(\sqrt{2x+4}+2)}{2\cancel{x}} = \lim_{x \rightarrow 0} \frac{\sqrt{2x+4}+2}{2} \\ &= \frac{\sqrt{2 \cdot 0 + 4} + 2}{2} = 2. \end{aligned}$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\sqrt{x} \cdot \sin(\sqrt{x})}{\sqrt{2x+4}-2} = \lim_{x \rightarrow 0} \frac{\sin(\sqrt{x})}{\sqrt{x}} \cdot \lim_{x \rightarrow 0} \frac{x}{\sqrt{2x+4}-2} = 1 \cdot 2 = \underline{2}.$$

b) We use the following transformation:

$$\lim_{x \rightarrow 0} \frac{x(e^{4x} - e^{2x})}{1 - \cos^3 x} = \lim_{x \rightarrow 0} \frac{e^{4x} - e^{2x}}{x} \cdot \frac{x^2}{1 - \cos^3 x} = \lim_{x \rightarrow 0} \frac{e^{4x} - e^{2x}}{x} \cdot \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^3 x}.$$

By the substitution $y = 2x \rightarrow 0$ if $x \rightarrow 0$ we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{4x} - e^{2x}}{x} &= \lim_{x \rightarrow 0} \frac{e^{2x}(e^{2x} - 1)}{x} = \lim_{x \rightarrow 0} e^{2x} \cdot \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = \lim_{x \rightarrow 0} e^{2x} \cdot \lim_{x \rightarrow 0} 2 \cdot \frac{e^{2x} - 1}{2x} \\ &= e^0 \cdot 2 \cdot \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \cdot 2 \cdot 1 = 2. \end{aligned}$$

On the other hand,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^3 x} &= \lim_{x \rightarrow 0} \frac{x^2}{(1 - \cos x)(1 + \cos x + \cos^2 x)} = \lim_{x \rightarrow 0} \frac{1}{\frac{1 - \cos x}{x^2}} \cdot \lim_{x \rightarrow 0} \frac{1}{1 + \cos x + \cos^2 x} \\ &= \frac{1}{1/2} \cdot \frac{1}{1 + 1 + 1} = \underline{\underline{\frac{2}{3}}}. \end{aligned}$$

Therefore

$$\lim_{x \rightarrow 0} \frac{x(e^{4x} - e^{2x})}{1 - \cos^3 x} = \lim_{x \rightarrow 0} \frac{e^{4x} - e^{2x}}{x} \cdot \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^3 x} = 2 \cdot \frac{2}{3} = \underline{\underline{\frac{4}{3}}}.$$